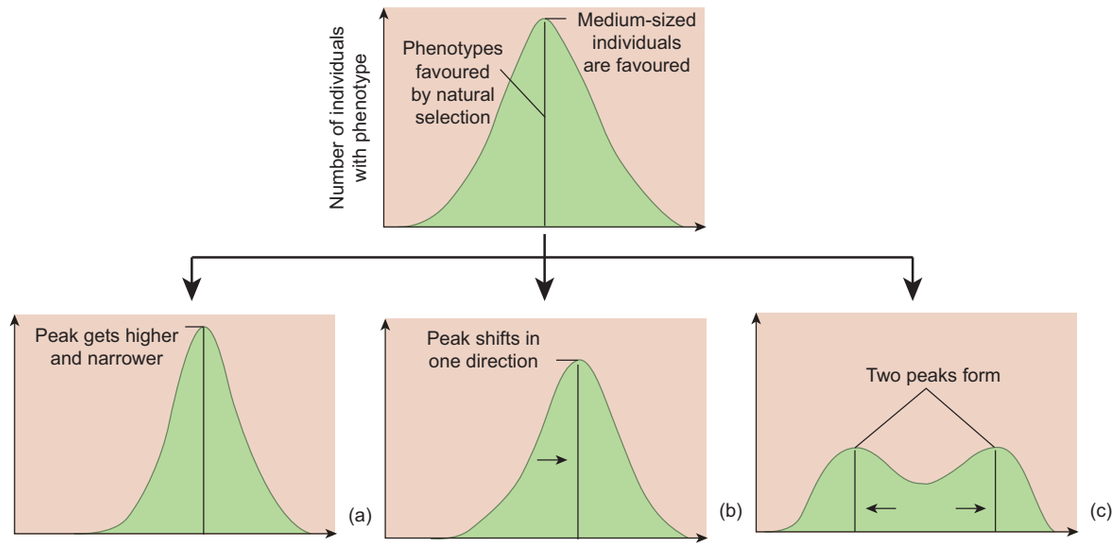


Fig 6.7 Operation of natural selection on different traits (a) Stabilising (b) Directional and (c) Disruptive

or more different species. It is also called adaptive radiation. E.g. Darwin's finches-beak size in relation to seed size inhabiting Galapagos islands.

Group selection and sexual selection are other types of selection. The two major group selections are Altrusim and Kin selection.

6.6.2 Gene flow

Movement of genes through gametes or movement of individuals in (immigration) and out (emigration) of a population is referred to as gene flow. Organisms and gametes that enter the population may have new alleles or may bring in existing alleles but in different proportions than those already in the population. Gene flow can be a strong agent of evolution **(Fig 6.8).**

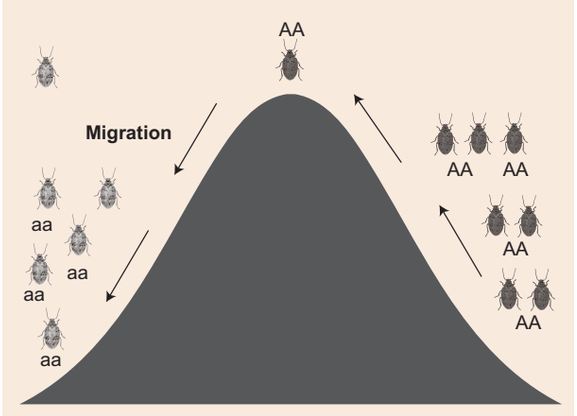


Fig 6.8 Gene flow

6.6.3 Genetic drift / Sewall Wright Effect

Genetic drift is a mechanism of evolution in which allele frequencies of a population change over generation due to chance (sampling error). Genetic drift occurs in all population sizes, but its effects are strong in a small population **(Fig. 6.9).** It may result in a loss of some alleles (including beneficial ones) and fixation of other alleles. Genetic drift can have major effects, when the population is reduced in size by natural disaster due to bottle neck effect or when a small group of population splits from the main population to form a new colony due to founder's effect.

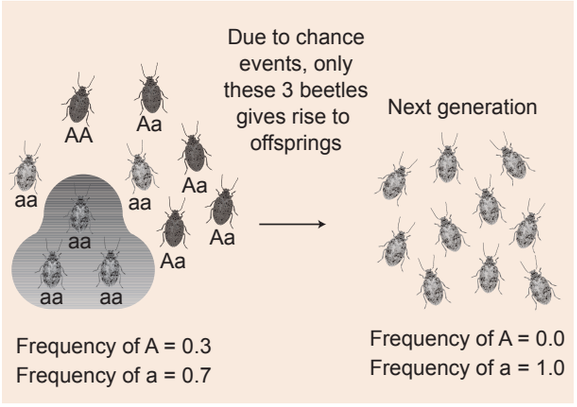


Fig 6.9 Genetic drift

XII Std Biology-Zoology Chapter-6 EM.indd 101 23/12/2022 12:25:51